Quantum Computing & Computational Chemistry

Foundations, Algorithms, and Practical Implementation

FOCI $\times$ Chemistry Workshop

Future of Computing Institute @ RPI

May 18, 2026

Workshop Outline

Hilbert Space, Kets, and Bras

The State Space of Quantum Mechanics

- A quantum state $|\psi\rangle$ resides in a complex **Hilbert space** $\mathcal{H}$, equipped with an inner product $\langle\varphi|\psi\rangle \in \mathbb{C}$.
- **Ket:** $|\psi\rangle \in \mathcal{H}$ represents a quantum state (as a column vector if $\mathcal{H}$ is finite dimensional).
  A wavefunction describes **probability amplitudes** which are **complex**.
  $\mathit{probability} = |\mathit{probability\ amplitude}|^2$
- **Bra:** $\langle\psi| \in \mathcal{H}^*$ is the dual vector (conjugate transpose), defined as $\langle\psi| = |\psi\rangle^\dagger$. $\dagger$ denotes a conjugate transposition.
- The inner product (bra-ket) $\langle\varphi|\psi\rangle$ quantifies overlap.
  Quantum states must be normalized: $\langle\psi|\psi\rangle = 1$.
  (Probabilities always sum to 1).

Physical Intuition: The bra-ket formalism abstracts away the representation (wavefunctions, spinors, or discrete vectors) and focuses on the linear algebraic structure. This abstraction is powerful for mapping continuous electronic wavefunctions to discrete qubits.

Wavefunctions $\neq$ Vectors: Wavefunctions are **always** normalized.

Operators, Observables, and Unitary Evolution

Acting on States

- A linear operator $\hat{O}$ acts on kets: $\hat{O}|\psi\rangle = |\psi'\rangle$.
  Matrix elements $\langle\varphi|\hat{O}|\psi\rangle$ are complex probability amplitudes.
- **Hermitian:** ($\hat{O}^\dagger = \hat{O}$) real eigenvalues and orthogonal eigenvectors.
  Hermitian operators represent **physical observables** (energy, spin, momentum).
  We can only measure real numbers in experiments.
- **Unitary operators** ($\hat{U}^\dagger \hat{U} = \hat{I}$) preserve norms: $\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle$.
  All quantum gates are unitary, ensuring probability conservation.
- The **outer product** $|\psi\rangle\langle\varphi|$ forms a rank-1 operator, frequently used as a projector onto a specific state or subspace.

Hermiticity guarantees measurable quantities, while unitarity guarantees reversible, norm-preserving dynamics.

One Qubit

The Computational Primitive

- A single qubit: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$.
  $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- **Single-Qubit Gates:** 2×2 unitary matrices.
  - Pauli gates: X (bit flip), Y, Z (phase flip)
    $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
  - Hadamard: $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ creates superposition
  - Rotations: $R_{x,y,z}(\theta) = e^{-i\theta P/2}$
    These are just exponentiations of the Pauli gates!

Superposition: $50 - 50$ probability of $|0\rangle$, $|1\rangle$

Convention: Qubits are always initialized as $|0\rangle$ unless specified.

Qiskit Implementation: One qubit

Applying a single gate.

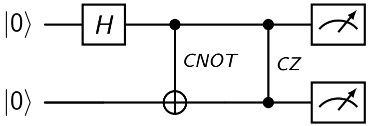
- **Qiskit: Q**uantum **I**nformation **S**cience **Kit**, an open-source library for handling quantum circuits and associated data structures.
- Qiskit's `QuantumCircuit` object directly mirrors the circuit model: qubits are indexed from 0 to $n-1$.
- By default, Qiskit uses little-endian bit ordering for state vectors, but circuit diagrams display qubit 0 at the top for visual consistency.

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(1) # Create a quantum circuit with one qubit
qc.h(0)                # Apply H gate to qubit 0
qc.measure_all()       # Measure qubits

qc.draw()              # Draw circuit
```

Two Qubits

- **Two Qubits:** State space of qubits A, B is a tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$ $\rightarrow 2 \times 2 = 4$ dimensional column vector.
- A general 2-qubit state: $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$.
- **Two-Qubit Gates:** State space of gates increases correspondingly.
- **Important Two-Qubit Gates:** CNOT and CZ introduce entanglement. CNOT flips the target iff control is $|1\rangle$, CZ imparts a phase of -1.

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \mathrm{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$



Note the two-qubit gate symbols! Why does the CZ gate look symmetric?

Qiskit Implementation: Two qubits

Applying two-qubit gates.

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2) #Create a quantum circuit with one qubit
qc.h(0)                # Apply H gate to qubit 0
qc.cx(0,1)             # Apply CNOT control 0, target 1
qc.cz(0,1)             # Apply CZ control 0, target 1
qc.measure_all()      # Measure qubits

qc.draw()              # Draw circuit
```

Basis Expansions, Tensor Products, and Entanglement

Representing States and Multi-Particle Systems

- Any state can be expanded in an orthonormal basis $\{|i\rangle\}$ satisfying $\langle i|j\rangle = \delta_{ij}$ and completeness $\sum_i |i\rangle\langle i| = \hat{I}$.
- Expansion: $|\psi\rangle = \sum_i c_i|i\rangle$, where $c_i = \langle i|\psi\rangle$. The probability of measuring outcome i is $|c_i|^2$.
- **Tensor Products:** For composite systems, the Hilbert space dimension multiplies: $\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = \dim(\mathcal{H}_A) \cdot \dim(\mathcal{H}_B)$.
- A general 2-qubit state: $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$.
- **Entanglement:** States that cannot be factored as $|\psi_A\rangle \otimes |\psi_B\rangle$ exhibit quantum correlations. Entanglement is the primary resource for quantum advantage in computational chemistry.

Measurement Postulate and Expectation Values

Extracting Information

- In quantum chemistry we usually are interested in **properties** instead of the wavefunction itself. Properties are extracted as **expectation values**.
- Measuring an observable $\hat{O} = \sum_k \lambda_k |k\rangle\langle k|$ on state $|\psi\rangle = \sum_k c_k |k\rangle$ yields eigenvalue λ_k with probability $p_k = |c_k|^2$.
- The post-measurement state collapses to $|k\rangle$ (destructive measurement).
- **Expectation Values:** $\langle \hat{O} \rangle_\psi = \langle \psi | \hat{O} | \psi \rangle = \sum_k \lambda_k p_k$.
  In practice, estimated over N shots with standard error $\sim 1/\sqrt{N}$.
- **Pauli Measurement Strategy:** Any Hermitian operator decomposes into Pauli strings $\hat{O} = \sum_k c_k P_k$.
  Each $P_k \in \{I, X, Y, Z\}^{\otimes n}$ is measured by rotating qubits to the Z-basis and taking the product of ± 1 outcomes.
- **Computational Chemistry:** The electronic Hamiltonian can be decomposed into $\mathcal{O}(N^4)$ Pauli terms.
  Grouping commuting terms and using randomized measurements drastically reduces the shot budget.

Measurement in the Circuit Model

The Measurement Postulate:

- Measuring an observable $\hat{O} = \sum_k \lambda_k |k\rangle\langle k|$ on state $|\psi\rangle$ yields eigenvalue λ_k with probability $p_k = |\langle k|\psi\rangle|^2$.
- Post-measurement, the state collapses to the corresponding eigenstate $|k\rangle$ (destructive process).
- In the computational basis, measuring qubit j in $\{|0\rangle, |1\rangle\}$ maps outcomes $0 \to +1$ and $1 \to -1$ for Pauli-Z expectation values.

Measurement Basis:

- The "meter" symbol denotes projective measurement in the Z-basis.
- To measure in the X- or Y-basis, rotations (H or $S^\dagger H$ respectively) are applied before the Z-measurement.
- Repeated shots yield a probability distribution; the expectation value is the statistical average over N samples with standard error $\sim 1/\sqrt{N}$.

Expectation Values and Pauli String Decomposition

- Hermitian operators (e.g., molecular Hamiltonian) decompose as:

$$\hat{O} = \sum_k c_k P_k, \quad P_k \in \{I, X, Y, Z\}^{\otimes n}, \quad c_k \in \mathbb{R}$$

- Expectation value becomes a weighted sum of Pauli expectations:

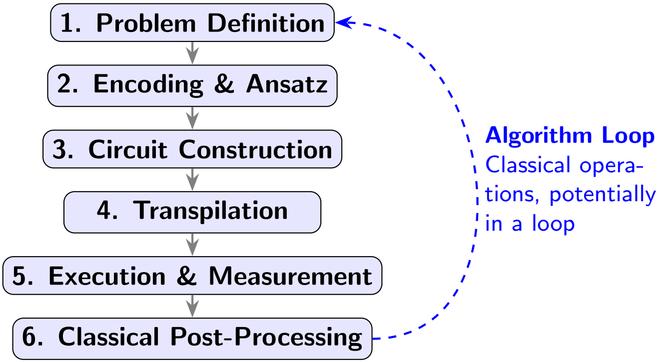
$$\langle \hat{O} \rangle_\psi = \sum_k c_k \langle P_k \rangle_\psi$$

- Each P_k has eigenvalues ± 1. Measuring $\langle P_k \rangle$ requires rotating qubits to the Z-basis, sampling bitstrings, and computing the parity-weighted average.

```
from qiskit.quantum_info import SparsePauliOp
```

- `SparsePauliOp` automatically groups commuting terms to reduce circuit depth and shot overhead.
- `StatevectorEstimator` computes exact $\langle \hat{O} \rangle$ on simulators; hardware uses `Estimator` with finite shots.
- For chemistry, $\hat{H}$ contains $\mathcal{O}(N^4)$ Pauli terms; efficient grouping and randomized measurement strategies are critical for tractable shot budgets.

Hybrid Quantum-Classical Workflow



- **Encoding:** Map problem (e.g., molecular Hamiltonian) to qubit operators via. fermion-to-qubit transformations (Jordan-Wigner, Bravyi-Kitaev).
- **Transpilation:** Decompose abstract circuits into hardware-native gates, optimize qubit routing, and apply compilation constraints.
- **Algorithm Loop:** Classical optimizer iteratively refines ansatz parameters until convergence (VQE, SQD, etc.).

Complexity Theory and Quantum Advantage

What Can Quantum Computers Actually Do?

- **Feynman's Argument (1981):** Simulating n spin-1/2 particles classically requires tracking 2^n amplitudes. At $n = 50$, this exceeds petabytes; time evolution demands exponential matrix operations.
- **Computability:** Quantum computers do *not* extend Turing-computability. They offer efficiency gains (polynomial or exponential speedups) for structured problems.
- **Complexity Classes:**
  - P: Deterministic polynomial time (classical)
  - BPP: Bounded probabilistic polynomial time (classical)
  - PSPACE: Polynomial space (classical)
  - BQP: Bounded-error quantum polynomial time. PSPACE $\supseteq$ BQP $\supseteq$ BPP
  - QMA: Quantum Merlin-Arthur. The local Hamiltonian problem is QMA-complete.
- **Practical Simulations:** Full Configuration Interaction is QMA-hard in the worst case, but real molecules exhibit structure. Heuristic methods (VQE, CC) exploit weak correlation; strongly correlated systems remain the true frontier for quantum advantage.

The Molecular Hamiltonian and Born-Oppenheimer Approximation

- Full non-relativistic molecular Hamiltonian (atomic units):

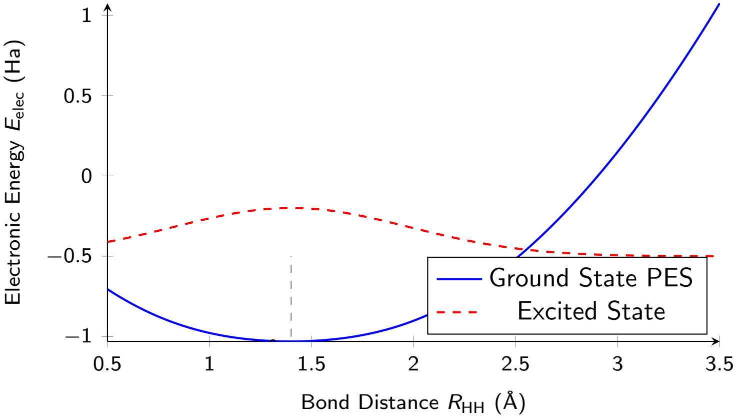
$$\hat{H} = \hat{T}_n + \hat{T}_e + \hat{V}_{nn} + \hat{V}_{ee} + \hat{V}_{ne}$$

- Nuclei are $\sim 1000\times$ heavier than electrons, $\sim 50\times$ slower
$\rightarrow$ **Born-Oppenheimer (BO) approximation**: treat nuclei as fixed point charges.

$$\hat{H}_{\text{elec}} = -\sum_i \frac{1}{2}\nabla_i^2 - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i<j} \frac{1}{r_{ij}} + \text{constants}$$

- Solving $\hat{H}_{\text{elec}}|\Psi_{\text{elec}}\rangle = E_{\text{elec}}|\Psi_{\text{elec}}\rangle$ yields the electronic energy and state.

Potential Energy Surface: H_2 Dissociation



Key Insight: The BO approximation reduces the full molecular problem to solving for $E_{\text{elec}}(R)$ at fixed nuclear geometries.

The Full Molecular Hamiltonian and Configuration Space

First-Quantized Form (Atomic Units)

$$\hat{H} = -\sum_{A=1}^{N_{\text{nuc}}} \frac{1}{2M_A}\nabla^2_{\mathbf{R}_A} - \sum_{i=1}^{N_{\text{el}}} \frac{1}{2}\nabla^2_{\mathbf{r}_i} + \sum_{A<B} \frac{Z_A Z_B}{R_{AB}} + \sum_{i<j} \frac{1}{r_{ij}} - \sum_{i,A} \frac{Z_A}{r_{iA}}$$

- **Spatial coordinates:** $\{\mathbf{R}_A\}_{A=1}^{N_{\text{nuc}}} \in \mathbb{R}^{3N_{\text{nuc}}}$ (nuclei) and $\{\mathbf{r}_i\}_{i=1}^{N_{\text{el}}} \in \mathbb{R}^{3N_{\text{el}}}$ (electrons)
- **Spin coordinates:** Each electron carries a discrete spin degree of freedom $\sigma_i \in \{\uparrow, \downarrow\}$, yielding a $2^{N_{\text{el}}}$-dimensional spin Hilbert space
- **Total wavefunction:** $\Psi(\mathbf{R}, \mathbf{r}, \boldsymbol{\sigma})$ depends on $3(N_{\text{nuc}} + N_{\text{el}})$ continuous variables and N_{el} discrete spin indices
- The number of dimensions grows with the number of electrons!
- Discretizing in space (e.g. a grid) leads to **exponentially** growing numbers of DOFs.
- Why is it called first-quantization?

The Antisymmetry Requirement for Fermions

Exchange Symmetry in Many-Electron Wavefunctions

- Electrons are spin-1/2 fermions. Quantum mechanics requires their total wavefunction to be **antisymmetric** under exchange of any two particles:

$$\Psi(\ldots, x_i, \ldots, x_j, \ldots) = -\Psi(\ldots, x_j, \ldots, x_i, \ldots)$$

  where $x_k = (\mathbf{r}_k, \sigma_k)$ denotes combined spatial and spin coordinates.
- Any function representing a solution (or approximate solution) must be antisymmetric.
- First-quantization: The burden of being antisymmetric is placed on the functions in the solution space (argument of the operator).
- Second-quantization: The burden of being antisymmetric is placed on the **operator** instead.
- Are qubit states on a quantum computer antisymmetric?

Basis Sets, LCAO, and the Hartree-Fock Method

Discretizing the Electronic Problem

- Instead of discretizing on a grid, use experimental observations and intuition to pick the basis.
- We can solve the equations for the hydrogen atom analytically (and that's pretty much the end).
- Main idea: Linear combination of atomic orbitals
  - Place orbitals centered on nuclei.
  - Linear combinations of atomic orbitals represent molecular orbitals.
- Solutions to the hydrogen atom are a starting point, but have a fundamental flaw: performing integrals on linear combinations becomes problematic!
- Well-established fix: Fit Gaussian functions to orbitals. Gaussians (and products of Gaussians) are easy to integrate.

Using Gaussians for Orbitals

Hierarchy:

- An orbital is represented as a linear combination of primitive Gaussians.
- Atomic orbitals are represented as a linear combination of orbitals.
- Molecular orbitals are represented as a linear combination of atomic orbitals.

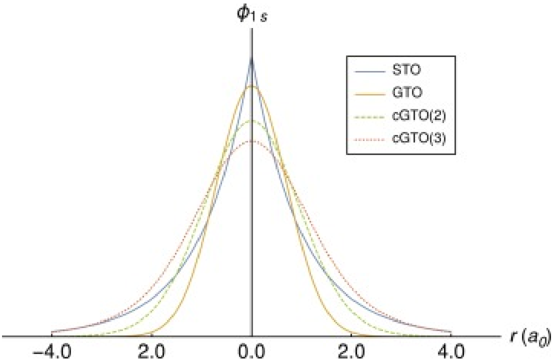
Primitive Gaussian-Type Orbital (GTO) Equation:

$$\chi_{abc}(\mathbf{r}) = N\, x^a y^b z^c\, e^{-\alpha|\mathbf{r}-\mathbf{R}|^2}$$

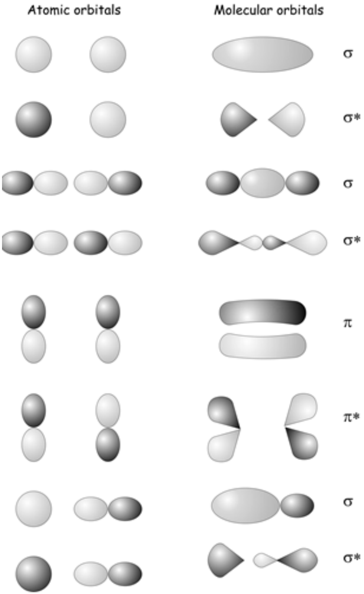
- $\mathbf{R}$: Center position (typically a nucleus)
- α: Exponent controlling the spatial decay/width
- a, b, c: Non-negative integers defining angular momentum $(s, p, d, f, \ldots)$
- N: Normalization constant ensuring $\int |\chi|^2 d\mathbf{r} = 1$

Gaussian Product Theorem: product of two Gaussians is another Gaussian, making electron repulsion integrals analytically tractable.

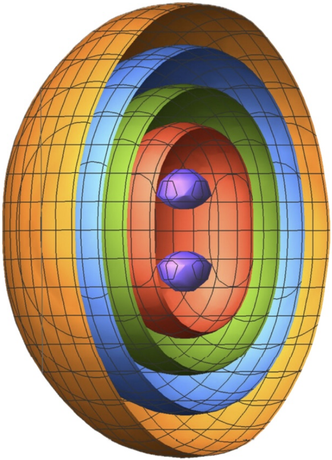
Gaussian Type Orbitals



Linear Combination of Atomic Orbitals



Grid Discretization



Workflow

- Pick your basis: https://www.basissetexchange.org/
- Define molecular geometry: (this can be an initial guess, and can iteratively converged by updating the Born-Oppenheimer model)
- Hartree Fock Algorithm
  - Each electron sees other electrons as a mean field (approximation)
  - The Roothaan-Hall equations form a generalized eigenvalue problem: $\mathbf{FC} = \mathbf{SC}\varepsilon$, solved self-consistently (SCF).
  - Output: Slater Determinant, a non-interacting (approximate) ground-state solution of a molecule.

Computational Chemistry Pipeline: Primitives → HF

- **Primitive Gaussians** (centered at atom A):

$$g_p(\mathbf{r}) = N_p\, x^l y^m z^n e^{-\alpha_p|\mathbf{r}-\mathbf{A}|^2}$$

- **Contracted Orbital Basis Functions** (fixed $d_{\mu p}$ from basissetexchange.org):

$$\chi_\mu(\mathbf{r}) = \sum_p d_{\mu p}\, g_p(\mathbf{r})$$

- **Molecular Orbitals** (LCAO, solve for $C_{\mu i}$) using Hartree Fock:

$$\psi_i(\mathbf{r}) = \sum_\mu C_{\mu i}\, \chi_\mu(\mathbf{r})$$

- **Slater Determinant** (N-electron wavefunction) from Hartree Fock:

$$|\Phi_0\rangle = \frac{1}{\sqrt{N!}} \det[\psi_i(\mathbf{x}_j)]$$

- **HF Solution** (Roothaan–Hall + SCF):

$$\mathbf{FC} = \mathbf{SC}\varepsilon$$

HF captures $\sim 99\%$ of total energy but completely misses **electron correlation** .

The Slater Determinant: Structure and Antisymmetry

Slater Determinant For N orthonormal spin-orbitals $\{\phi_i\}$, the many-electron wavefunction is:

$$|\Phi_0\rangle(x_1, \dots, x_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(x_1) & \phi_1(x_2) & \cdots & \phi_1(x_N) \\ \phi_2(x_1) & \phi_2(x_2) & \cdots & \phi_2(x_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_N(x_1) & \phi_N(x_2) & \cdots & \phi_N(x_N) \end{vmatrix}$$

Antisymmetry by Construction Swapping any two electron coordinates (e.g., $x_1 \leftrightarrow x_2$) swaps two columns of the matrix:

$$|\Phi_0\rangle(\dots, x_2, \dots, x_1, \dots) = -|\Phi_0\rangle(\dots, x_1, \dots, x_2, \dots)$$

The Antisymmetry Requirement for Fermions

Exchange Symmetry in Many-Electron Wavefunctions

- Electrons are spin-1/2 fermions. Quantum mechanics requires their total wavefunction to be **antisymmetric** under exchange of any two particles:

$$\Psi(\dots, x_i, \dots, x_j, \dots) = -\Psi(\dots, x_j, \dots, x_i, \dots)$$

  where $x_k = (\mathbf{r}_k, \sigma_k)$ denotes combined spatial and spin coordinates.
- A naive product of orbitals (Hartree product) $\Phi = \phi_1(x_1)\phi_2(x_2)\cdots\phi_N(x_N)$ is symmetric under exchange and violates fermionic statistics.
- If two electrons occupy the same quantum state ($x_i = x_j$), antisymmetry forces $\Psi = 0$.
  This is the mathematical origin of the **Pauli exclusion principle**.
- A **Slater determinant** enforces this automatically and satisfies the Pauli exclusion principle. Swapping two electrons will induce the -ve sign.
- **Occupation Number Representation:** Each spin-orbital is labeled $1, \dots, 2K$. Occupation number representations map to a bitstring $|n_1 n_2 \dots n_{2K}\rangle$, where $n_i \in \{0, 1\}$.
  This maps directly to qubits and is not antisymmetric.

Fock Space

Fock space: the complete vector space spanned by all possible Slater determinants constructed from a finite orbital basis.

- **Reference State:** The Hartree-Fock ground-state determinant $|\Phi_0\rangle$.
- Exciting electrons from occupied to virtual orbitals generates single, double, triple, etc., excitations: $|\Phi_i^a\rangle, |\Phi_{ij}^{ab}\rangle, \ldots$
- For N electrons distributed among $2K$ spin-orbitals, the number of unique determinants is $\binom{2K}{N}$. This scales combinatorially with the number of electrons and orbitals.
- The exact correlated wavefunction (within the limit of your discretization) lies in this space: $|\Psi_{\text{exact}}\rangle = c_0|\Phi_0\rangle + \sum c_i^a|\Phi_i^a\rangle + \sum c_{ij}^{ab}|\Phi_{ij}^{ab}\rangle + \ldots$
- Classical solvers must truncate the expansion (CISD, CCSD), sacrificing accuracy for computational feasibility.
- Quantum computers map Fock space determinants directly to computational basis states, enabling systematic exploration toward the **Full CI** (FCI) limit without explicit truncation.

Fock Space Structure: The Hamiltonian Matrix

The FCI Problem as a Linear Algebra Problem The full Hamiltonian in the Fock space basis is a massive matrix whose rows and columns correspond to Slater determinants:

$$H = \begin{pmatrix} \langle\Phi_0|\hat{H}|\Phi_0\rangle & \langle\Phi_0|\hat{H}|\Phi_i^a\rangle & \langle\Phi_0|\hat{H}|\Phi_{ij}^{ab}\rangle & \cdots \\ \langle\Phi_i^a|\hat{H}|\Phi_0\rangle & \langle\Phi_i^a|\hat{H}|\Phi_i^a\rangle & \langle\Phi_i^a|\hat{H}|\Phi_{ij}^{ab}\rangle & \cdots \\ \langle\Phi_{ij}^{ab}|\hat{H}|\Phi_0\rangle & \langle\Phi_{ij}^{ab}|\hat{H}|\Phi_i^a\rangle & \langle\Phi_{ij}^{ab}|\hat{H}|\Phi_{ij}^{ab}\rangle & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Key Points:

- The exact ground state $|\Psi_0\rangle$ is the eigenvector corresponding to the lowest eigenvalue E_0.
- **Slater-Condon Rules:** The matrix is extremely sparse. Elements are zero if determinants differ by > 2 spin-orbitals.

Computational Chemistry Pipeline: Primitives → HF → FCI

- **Primitive Gaussians** (centered at atom A):

$$g_p(\mathbf{r}) = N_p \, x^l y^m z^n e^{-\alpha_p |\mathbf{r}-\mathbf{A}|^2}$$

- **Contracted Orbital Basis Functions** (fixed $d_{\mu p}$):

$$\chi_\mu(\mathbf{r}) = \sum_p d_{\mu p} \, g_p(\mathbf{r})$$

- **Molecular Orbitals** (LCAO, solve for $C_{\mu i}$ using Hartree Fock):

$$\psi_i(\mathbf{r}) = \sum_\mu C_{\mu i} \, \chi_\mu(\mathbf{r})$$

- **Slater Determinant** (N-electron reference) from Hartree Fock:

$$|\Phi_0\rangle = \frac{1}{\sqrt{N!}} \det[\psi_i(\mathbf{x}_j)]$$

- **Full CI Expansion** (exact wavefunction in Fock space, up to discretization):

$$|\Psi_{\mathrm{FCI}}\rangle = c_0|\Phi_0\rangle + \sum_{i,a} c_i^a |\Phi_i^a\rangle + \sum_{i<j,a<b} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \cdots$$

Slater Determinants and Second Quantization

- **Occupation Number Representation:** Each spin-orbital is labeled $1, \ldots, 2K$. A determinant maps to a bitstring $|n_1 n_2 \ldots n_{2K}\rangle$, where $n_i \in \{0, 1\}$. This maps directly to qubits.
  Occupation number vectors do **not** enforce antisymmetry. This requires the transition to second quantization.
- **Second Quantization:** Instead of antisymmetrizing wavefunctions, we encode antisymmetry into operator algebra:

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \quad \{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0$$

- The Hamiltonian becomes:

$$\hat{H} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} g_{pqrs} a_p^\dagger a_q^\dagger a_r a_s$$

- **Slater-Condon Rules:** Matrix elements between determinants are non-zero only if they differ by ≤ 2 spin-orbitals. This sparsity is exploited in both classical and quantum solvers.
- The creation and annihilation operators $a_p^\dagger, a_p$ can be represented as matrices $\rightarrow$ we are getting closer to making the connection with quantum computers!

Second-Quantized Molecular Hamiltonian

$$\hat{H} = \sum_{p,q} h_{pq}\, a_p^\dagger a_q + \frac{1}{2} \sum_{p,q,r,s} g_{pqrs}\, a_p^\dagger a_q^\dagger a_r a_s$$

where $a_p^\dagger, a_p$ create/annihilate electrons in spin-orbital p, and sums run over all $2K$ spin-orbitals.

- **One-electron** (kinetic + nuclear attraction):

$$h_{pq} = \int \chi_p^*(\mathbf{r}) \left[-\frac{1}{2}\nabla^2 - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \right] \chi_q(\mathbf{r})\, d\mathbf{r}$$

- **Two-electron** (Coulomb repulsion):

$$g_{pqrs} = \iint \chi_p^*(\mathbf{r}_1)\chi_q^*(\mathbf{r}_2) \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \chi_r(\mathbf{r}_1)\chi_s(\mathbf{r}_2)\, d\mathbf{r}_1 d\mathbf{r}_2$$

Key Properties:

- Real basis $\Rightarrow$ Symmetry relations $h_{pq} = h_{qp}$, $g_{pqrs} = g_{qpsr} = g_{rspq} = g_{srpq}$
- Scales as $\mathcal{O}(N^4)$ with basis size N; forms the core data structure for both classical and quantum solvers
- Once you have a HF solution, you can form the integral tensors.

Second Quantization Operators, Jordan-Wigner Mapping

In the occupation basis $\{|0\rangle, |1\rangle\}$ for orbital j:

$$a_j^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad a_j = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

- **Action on states:** $a_j^\dagger|0\rangle = |1\rangle, \quad a_j|1\rangle = |0\rangle$
- $a_j|0\rangle = 0, a_j^\dagger|1\rangle = 0$: annihilating a non-existent electron yields zero

Pauli Decomposition Expressing the 2×2 matrices in terms of Pauli operators:

$$a_j = \frac{X_j + iY_j}{2}, \quad a_j^\dagger = \frac{X_j - iY_j}{2}$$

Jordan-Wigner Transformation To preserve fermionic anticommutation across all orbitals, we attach a parity string of Z operators:

$$a_j^\dagger = \left(\bigotimes_{i<j} Z_i \right) \otimes \frac{X_j - iY_j}{2}, \quad a_j = \left(\bigotimes_{i<j} Z_i \right) \otimes \frac{X_j + iY_j}{2}$$

The Z-string enforces fermionic parity. This mapping converts the second-quantized Hamiltonian into a sum of Pauli strings.

Fermion-to-Qubit Mapping and the VQE Algorithm

Translating Chemistry to Quantum Circuits

- **Jordan-Wigner (JW) Transformation:** Maps fermionic operators to Pauli strings while preserving anticommutation:

$$a_j^\dagger = \left(\bigotimes_{i<j} Z_i \right) \otimes \frac{X_j - iY_j}{2}$$

  The Hamiltonian becomes a sum of Pauli strings $\hat{H} = \sum_k c_k P_k$.

- **Variational Quantum Eigensolver (VQE):**
  1. Prepare parametrized ansatz $|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})|\Phi_0\rangle$ on hardware.
  2. Measure $\langle \hat{H} \rangle = \sum_k c_k \langle P_k \rangle$ via repeated shots.
  3. Classical optimizer updates $\boldsymbol{\theta}$ to minimize energy (variational principle guarantees $E(\boldsymbol{\theta}) \geq E_0$).
  4. Iterate until convergence.

- **Challenges:** Barren plateaus (vanishing gradients), shot noise, hardware decoherence, and local minima in the optimization landscape.

Building the VQE Ansatz

Initial State: The Hartree–Fock reference $|\Phi_0\rangle$ maps directly to a computational basis state (e.g., $|1100\ldots\rangle$ for occupied spin-orbitals).

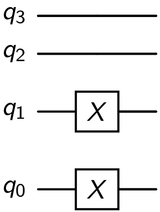
Unitary Coupled Cluster (UCC) Ansatz:

$$|\psi(\boldsymbol{\theta})\rangle = \exp\left(\sum_k \theta_k(\hat{T}_k - \hat{T}_k^\dagger)\right)|\Phi_0\rangle$$

- **Singles:** $\hat{T}_1 = a_a^\dagger a_i$ (promote electron $i \to a$)
- **Doubles:** $\hat{T}_2 = a_a^\dagger a_b^\dagger a_j a_i$ (promote electrons $i, j \to a, b$)
- Unitarity ($\hat{U}^\dagger \hat{U} = \hat{I}$) guarantees norm preservation and variational stability.
- Each exponential $\exp(-i\theta_k(\hat{T}_k - \hat{T}_k^\dagger))$ is approximated via Trotterization into sequences of CNOT gates and single-qubit rotations.
- θ_k are the excitation amplitudes (parameters to be optimized).

Circuit for VQE Ansatz: Hartree–Fock Initial State

Mapping to Qubits: For H_2 in STO-3G, we have 2 spatial orbitals $\Rightarrow$ 4 spin-orbitals $\Rightarrow$ 4 qubits.
Preparation Circuit: The HF state $|\Phi_0\rangle = |1100\rangle$ is prepared by applying X gates to the occupied qubits:



Note:

- Qiskit Nature uses $\alpha_0, \beta_0, \alpha_1, \beta_1$ ordering by default.
- Any valid Fock Space Slater Determinant will satisfy:

$$\#\ of\ ones = \#\ of\ electrons$$

Circuit for VQE Ansatz: UCCSD Trotterization

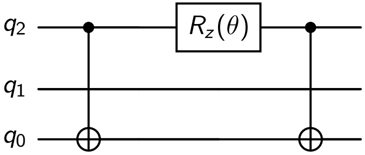
Excitation Operators → Pauli Strings:

- Single: $\hat{T}_1 = a_a^\dagger a_i \xrightarrow{\text{JW}} \frac{1}{2}(X_i X_a + Y_i Y_a) + \text{parity terms}$
- Double: $\hat{T}_2 = a_a^\dagger a_b^\dagger a_j a_i \xrightarrow{\text{JW}}$ 4-qubit Pauli strings

Trotterized Unitary:

$$e^{-i\theta(\hat{T}-\hat{T}^\dagger)} \approx e^{-i\theta P_1} e^{-i\theta P_2} \dots$$

Rough Circuit Structure (Single Excitation $0 \to 2$):



Key Points:

- CNOT ladders compute the parity of the involved qubits.
  This is very expensive!
- Double excitations follow the same pattern with deeper CNOT chains.

Spectrum of Quantum Chemistry Algorithms

Quantum Algorithm Landscape:

Variational / NISQ

- VQE (UCC, Hardware-Efficient)
- SQD / Subspace Expansion
- ADAPT-VQE
- VQMC

Phase Estimation / FT

- QPE (Quantum Phase Estimation)
- Hadamard Test
- QCELS / QMEGS
- Krylov / ODMD
- QITE (Quantum Imaginary Time Evolution)

Selection Criteria: Circuit depth, coherence time, error tolerance, sampling overhead, and target precision dictate the algorithm choice.

Classical Correlation Methods and Unitary Coupled Cluster

Beyond Hartree-Fock

- **Full CI (FCI):** Expands $|\Psi\rangle = \sum_I c_I |\Phi_I\rangle$ over all determinants. Exact within the basis, but scales combinatorially $\binom{2K}{N}$, limiting use to ~ 20 electrons.
- **Truncated CI (CISD):** Includes all singly and doubly excited determinants. Captures $\sim 90 - 95\%$ of correlation but lacks size-consistency.
- **Coupled Cluster (CC):** $|\Psi_{\text{CC}}\rangle = e^{\hat{T}}|\Phi_0\rangle$, where $\hat{T} = \hat{T}_1 + \hat{T}_2 + \dots$ are single, double $\dots$ excitation operators.
  CCSD is size-consistent and scales as $\mathcal{O}(N^6)$.
  CCSD(T) is the classical "gold standard".
- **Unitary Coupled Cluster (UCC):** $|\Psi_{\text{UCC}}\rangle = e^{\hat{T}-\hat{T}^\dagger}|\Phi_0\rangle$.
  Unitarity ensures norm preservation, making it suitable for quantum circuits. The Baker-Campbell-Hausdorff expansion does not terminate, making this intractable in general classically.
- **Strong Correlation:** Bond breaking and transition metals exhibit near-degenerate determinants. Single-reference methods fail; quantum computers target the FCI solution directly.

Summary and Outlook

- Quantum computing offers a natural mapping to fermionic Hilbert spaces, bypassing the exponential scaling of classical wavefunction methods.
- The Jordan-Wigner transformation and second quantization bridge electronic structure theory and qubit operations.
- VQE and SQD provide noise-resilient pathways for NISQ devices, while QPE promises asymptotic advantage on fault-tolerant hardware.
- Enforcing antisymmetry for electrons dominates the quantum computational overhead.
- **Open Challenges:** Error mitigation, ansatz design avoiding barren plateaus, efficient Pauli grouping, and demonstrating practical quantum advantage for chemically relevant systems.
- The intersection of quantum information theory and computational chemistry is rapidly evolving, with hybrid classical-quantum algorithms leading the way.

Topics Not Covered

- Circuit transpilation, SWAP routing, and hardware topology constraints
- NISQ noise models, error mitigation (ZNE, PEC, VFC), and error correction
- Fault-tolerant architectures, logical qubits, and surface codes
- Alternative mappings: Bravyi-Kitaev, Parity, BKSF
- HPC-Quantum co-processing and hybrid workflows at RPI
- Lattice models (Hubbard, Heisenberg) and condensed matter applications
- Excited states: LR-VQE, QSE, QPE with ancilla, and state preparation
- Gradient computation: Parameter-shift rule, stochastic estimation, barren plateaus
- Software ecosystems: OpenFermion, Tequila, PennyLane, Qiskit internals
- Benchmarking, quantum volume, and practical advantage thresholds